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Microsoft

Microsoft Azure Virtual Training Day: AI Fundamentals

About this course

Course objectives:

- Describe Artificial Intelligence workloads and considerations
- Describe fundamental principles of machine learning on Azure
- Describe features of computer vision workloads on Azure
- Describe features of Natural Language Processing (NLP) workloads on Azure

This course is supplemented by online training at https://aka.ms/AzureLearn_AIFundamentals

Course Agenda

Module 1: Explore fundamentals of Artificial Intelligence

- Introduction to Artificial Intelligence
- Artificial Intelligence in Microsoft Azure

Module 2: Explore fundamentals of machine learning

- Introduction to Machine Learning
- Azure Machine Learning

Module 3: Explore fundamentals of computer vision

- Computer Vision Concepts
- Creating Computer Vision solutions in Azure

Module 4: Explore fundamentals natural language processing

- Introduction to Natural Language Processing
- Building Natural Language Solutions in Azure

Demos

- Demos in this course are based on exercises in Microsoft Learn

Module 1:

Explore Fundamentals of Artificial Intelligence

- Lesson 1: Introduction to Artificial Intelligence
- Lesson 2: Artificial Intelligence in Microsoft Azure

Lesson 1: Introduction to Artificial Intelligence

What is Artificial Intelligence?

Software that imitates human capabilities

- Predicting outcomes and recognizing patterns based on historic data
- Recognizing abnormal events and making decisions
- Interpreting visual input
- Understanding language, and engaging in conversations
- Extracting information from sources to gain knowledge

Common Artificial Intelligence Workloads

Machine Learning	Predictive models based on data and statistics – the foundation for AI
Anomaly Detection	Systems that detect unusual patterns or events, enabling pre-emptive action
Computer Vision	Applications that interpret visual input from cameras, images, or videos
Natural Language Processing	Applications that can interpret written or spoken language, and engage in dialogs with human users
Knowledge Mining	Extract information from data sources to create a searchable knowledge store

Principles of Responsible AI

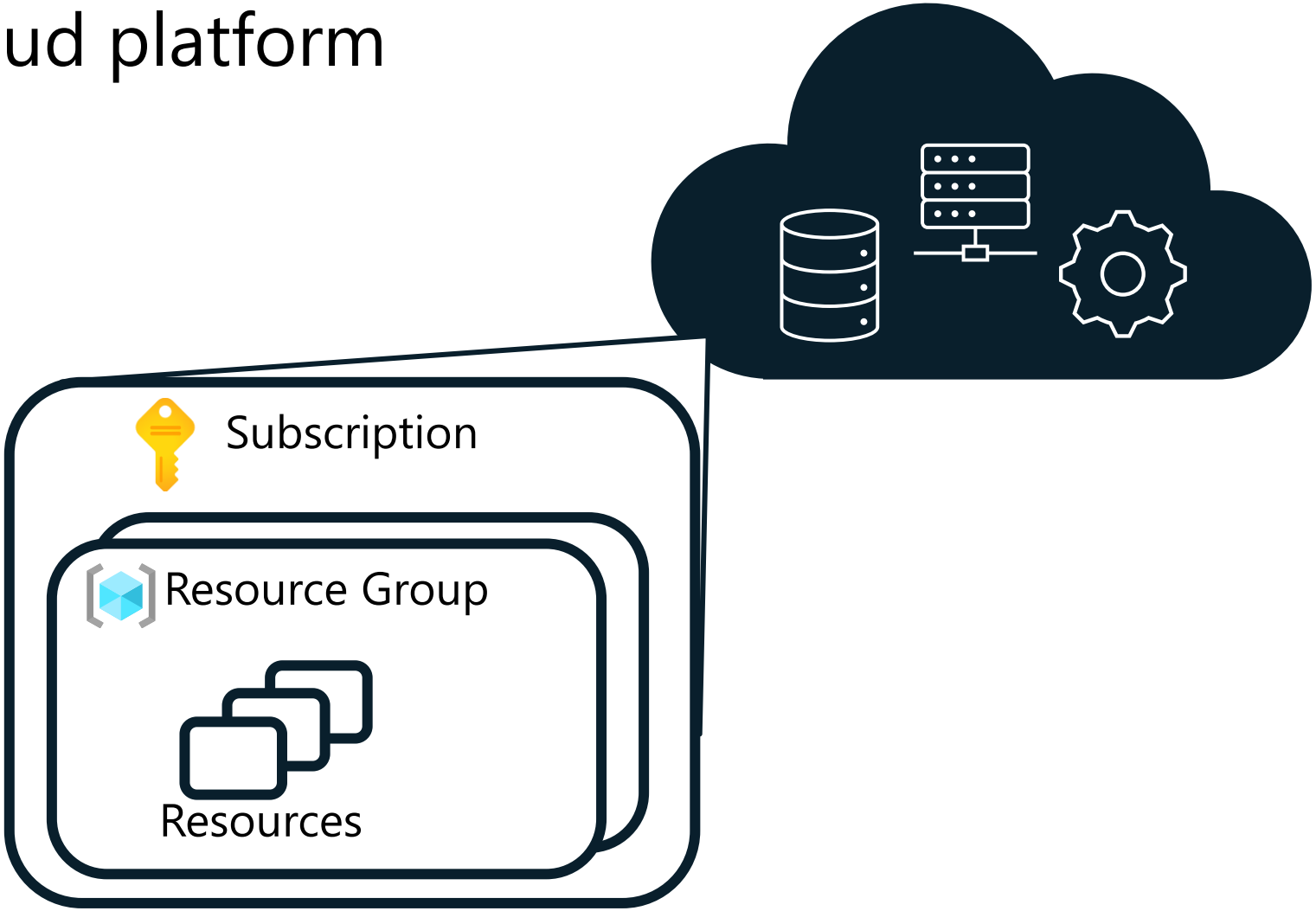
	Challenge or Risk	Example
Fairness	Bias can affect results	A loan-approval model discriminates by gender due to bias in the data with which it was trained
Reliability & Safety	Errors may cause harm	An autonomous vehicle experiences a system failure and causes a collision
Privacy & Security	Data could be exposed	A medical diagnostic bot is trained using sensitive patient data, which is stored insecurely
Inclusiveness	Solutions may not work for everyone	A predictive app provides no audio output for visually impaired users
Transparency	Users must trust a complex system	An AI-based financial tool makes investment recommendations - what are they based on?
Accountability	Who's liable for AI-driven decisions?	An innocent person is convicted of a crime based on evidence from facial recognition – who's responsible?

Lesson 2: Artificial Intelligence in Microsoft Azure

Azure Basics

Scalable, reliable cloud platform

- Data storage
- Compute
- Services

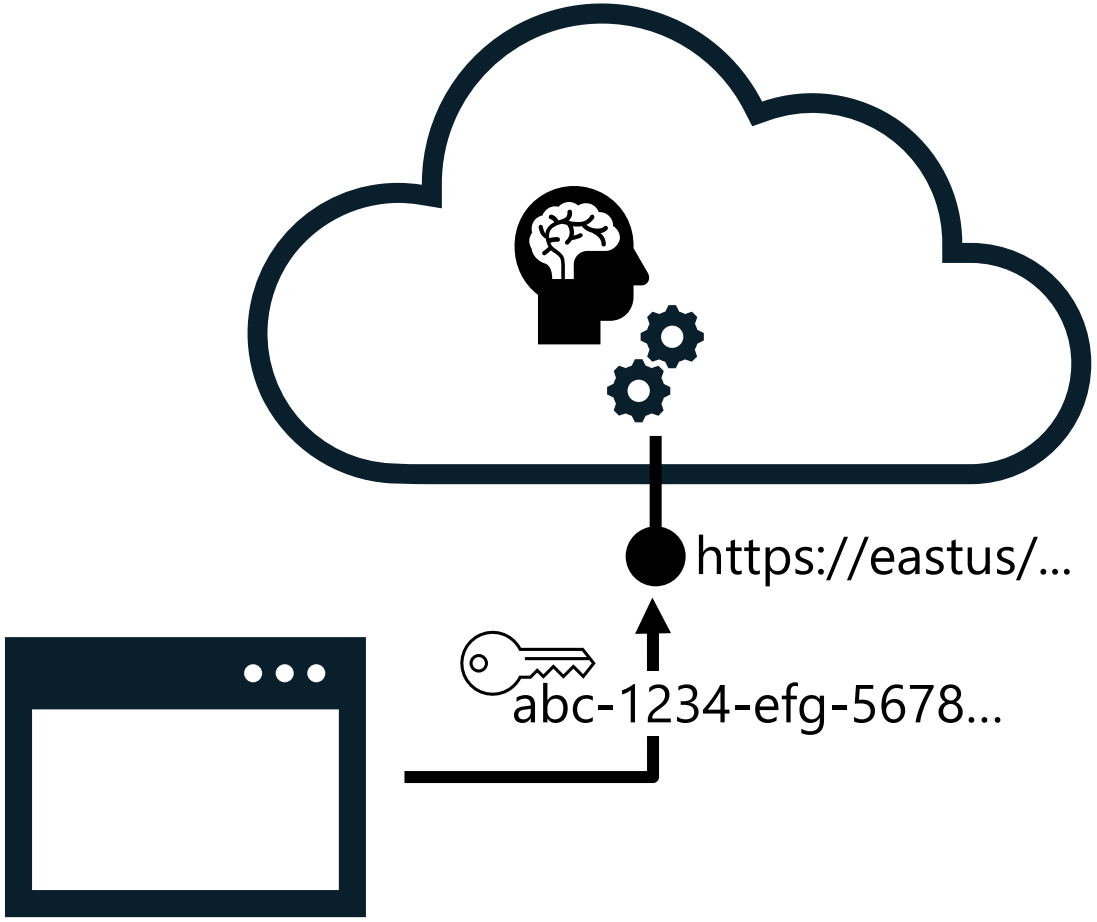


AI Services in Microsoft Azure

Azure Machine Learning	A platform for training, deploying, and managing machine learning models
Cognitive Services	A suite of services with four main pillars: Vision, Speech, Language, Decision
Azure Bot Service	A cloud-based platform for developing and managing conversational bots
Azure Cognitive Search	Data extraction, enrichment, and indexing for intelligent search and knowledge mining

Cognitive Services

- AI application resources in an Azure subscription:
  - Standalone resources for specific services
  - General *Cognitive Services* resource for multiple services
- Consumed by applications via:
  - A REST endpoint (https:// address)
  - An authentication key or authorization token



Demo

Explore Cognitive Services

Module 2:

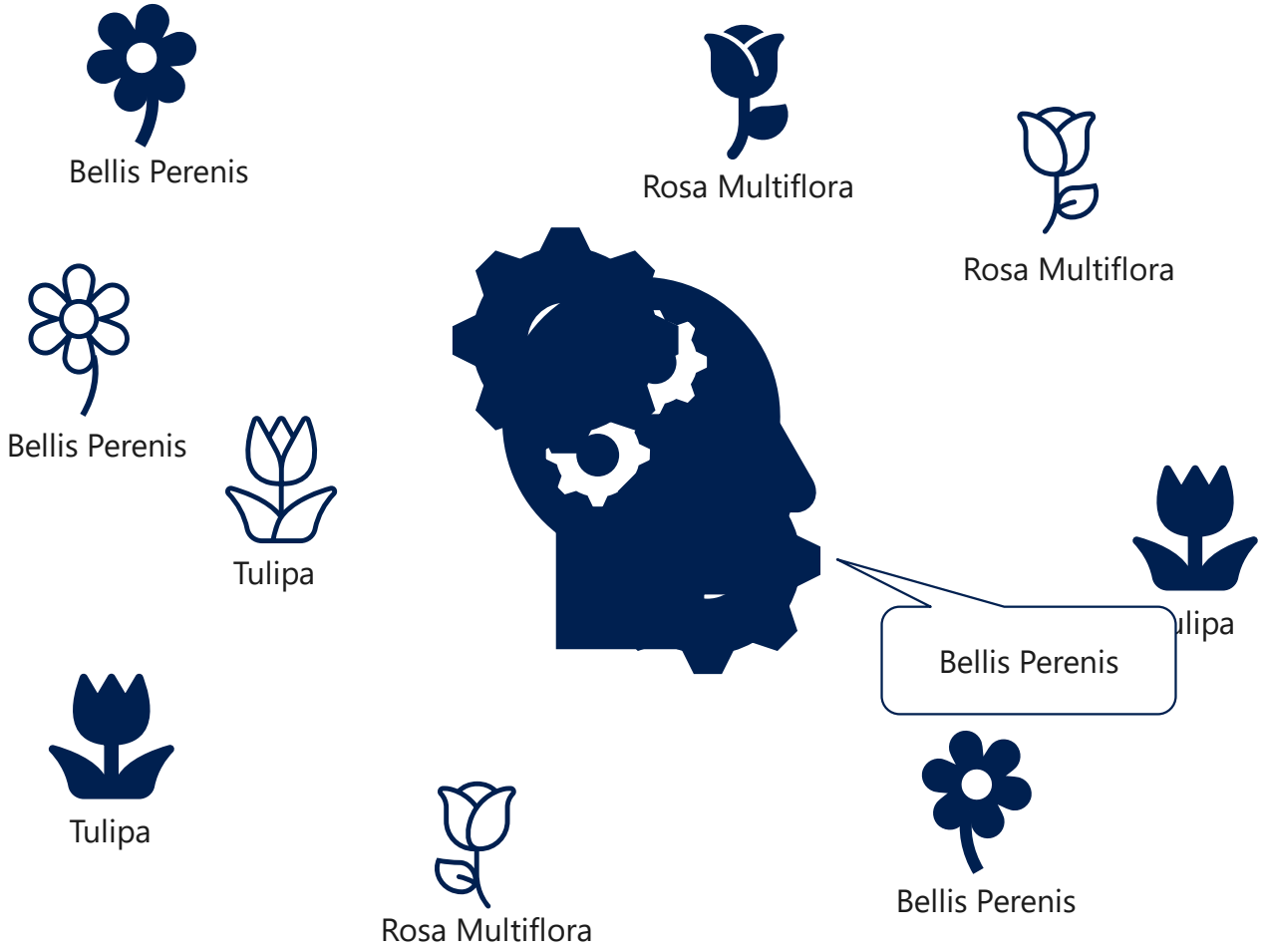
Explore Fundamentals of Machine Learning

- Lesson 1: Introduction to Machine Learning
- Lesson 2: Azure Machine Learning

Lesson 1: Introduction to Machine Learning

What is Machine Learning?

Creating predictive models by finding relationships in data



1. A botanist collects some samples of flowers
2. Each sample has a set of *features* (characteristics) and a *label* (the species)
3. An algorithm is used to find the relationship between the features and the label
4. The result is a *model* that encapsulates those relationships
5. The model can predict the label of a new sample based on its features

Types of Machine Learning

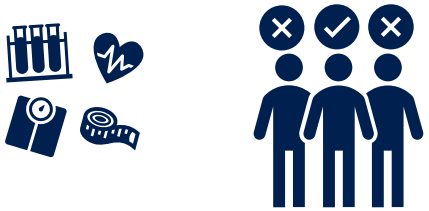
Machine Learning

Supervised Machine Learning
Training data includes known labels

Regression
Label is a numeric value

Predict the number of bike rentals based on day, season, and weather

Classification
Label is a categorization (or *class*)



Predict whether a patient is at-risk for diabetes based on clinical measurements

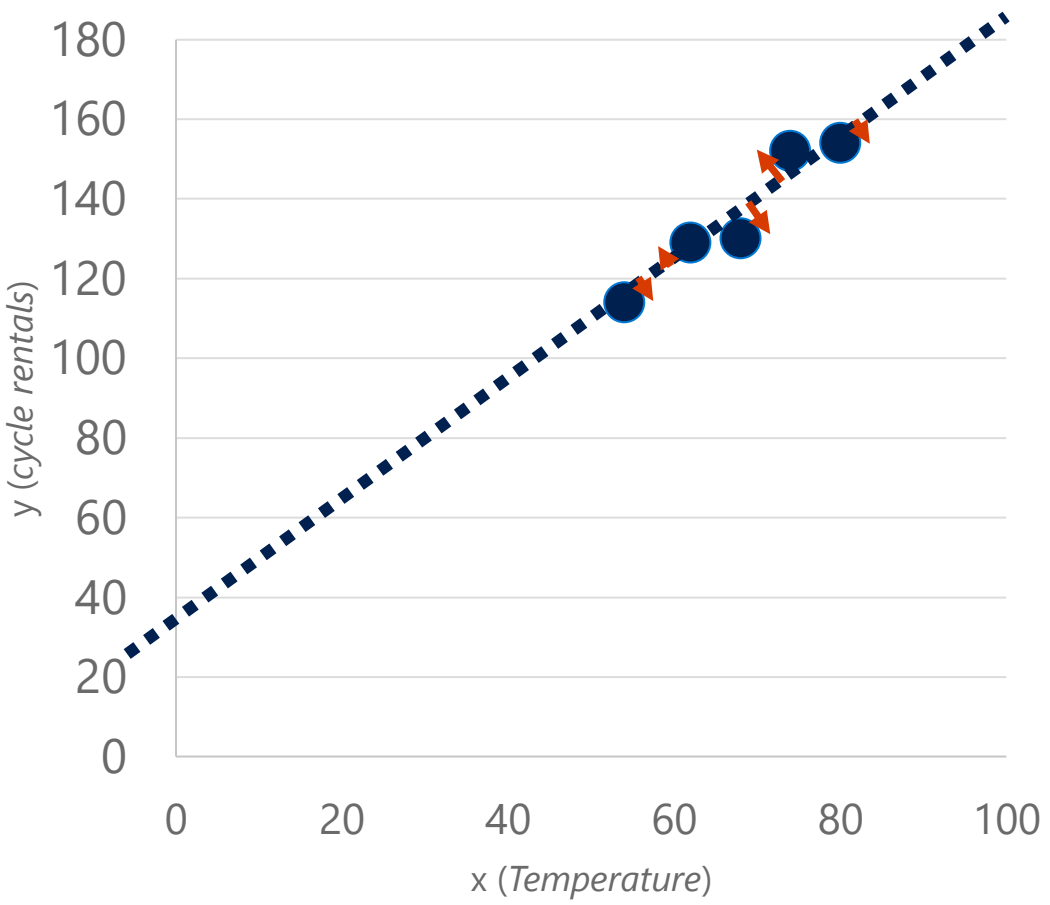
Unsupervised Machine Learning
Training data is unlabeled

Clustering
Similar items are grouped together

Vehicles with similar emissions and fuel efficiency characteristics are separated into clusters

Regression

	x	y
Training	56	115
	61	126
	67	137
	72	140
	76	152
	82	156
Validation	54	114
	62	129
	68	130
	74	152
	80	154

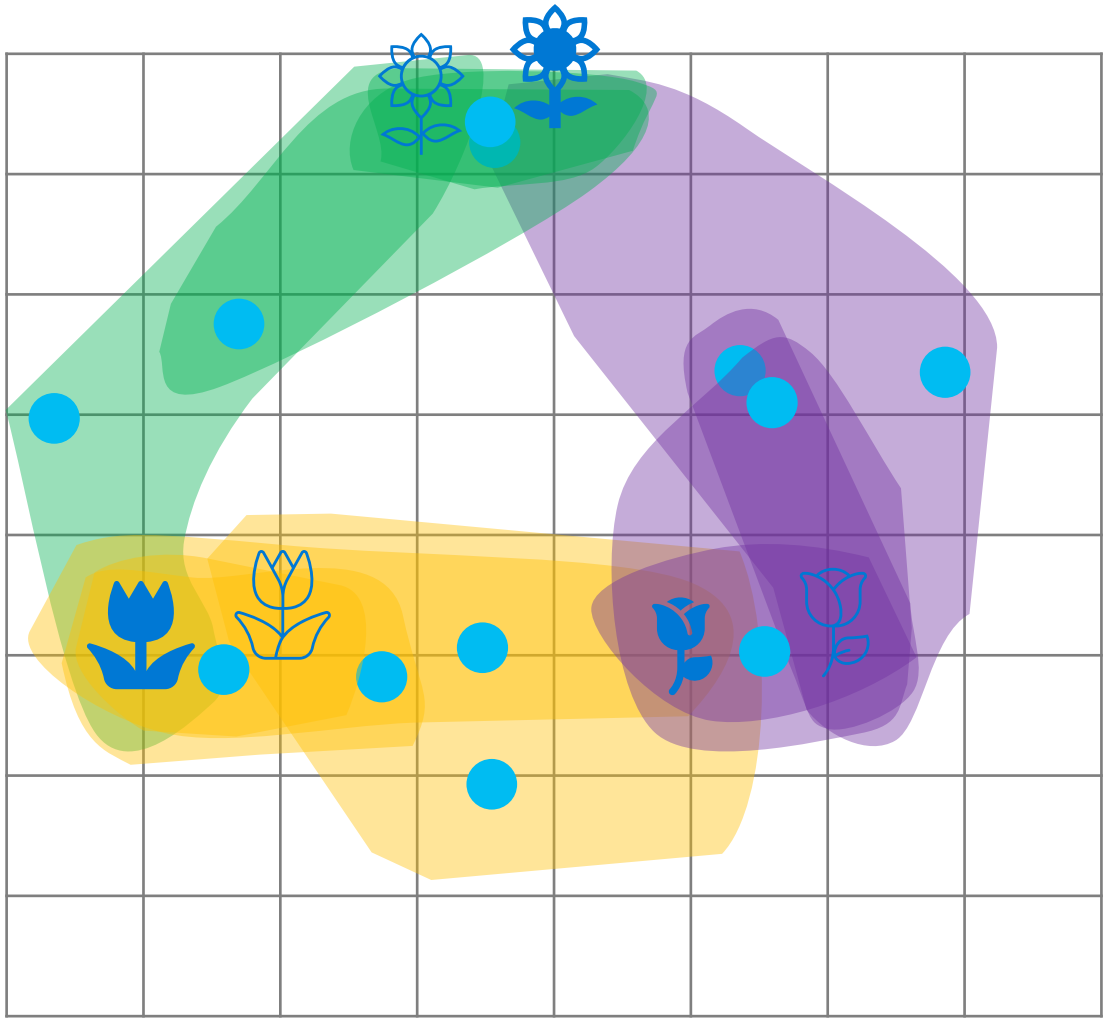


Classification

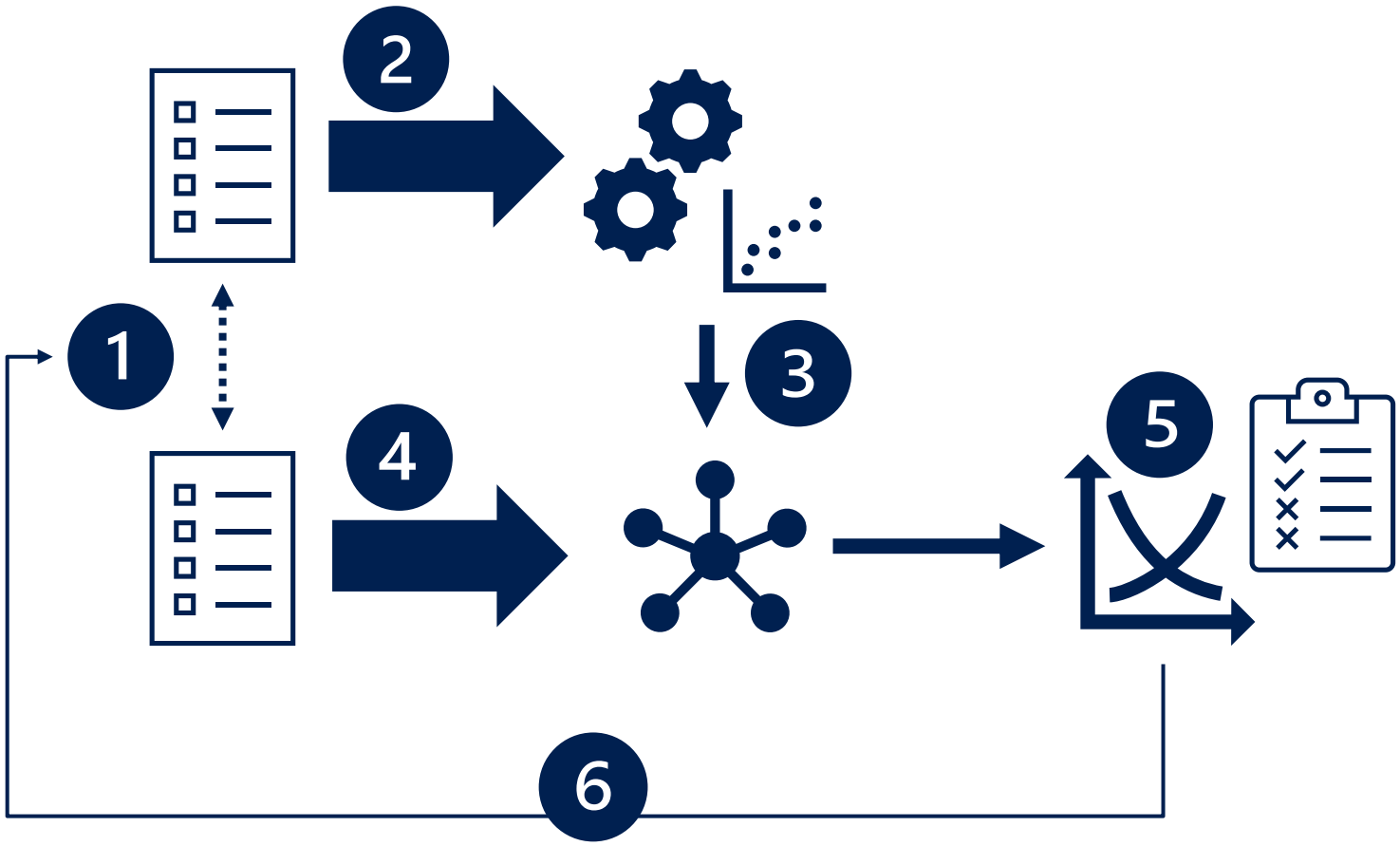
	x	y
Training	[4,2,3.2,10]	0
	[6,3,1.7,12]	1
	[5,2,3.5,11]	0
	[4,1,2.9,10]	0
	[7,4,2.1,11]	1
Validation	[3,1,2.8,10]	0
	[7,3,1.8,11]	1
	[4,8,2.5,10]	0
	[4,1,3,1,11]	1

Clustering

	6	3
	5	3
	2	3
	1	3
	3	8
	4	8



Model Training and Validation

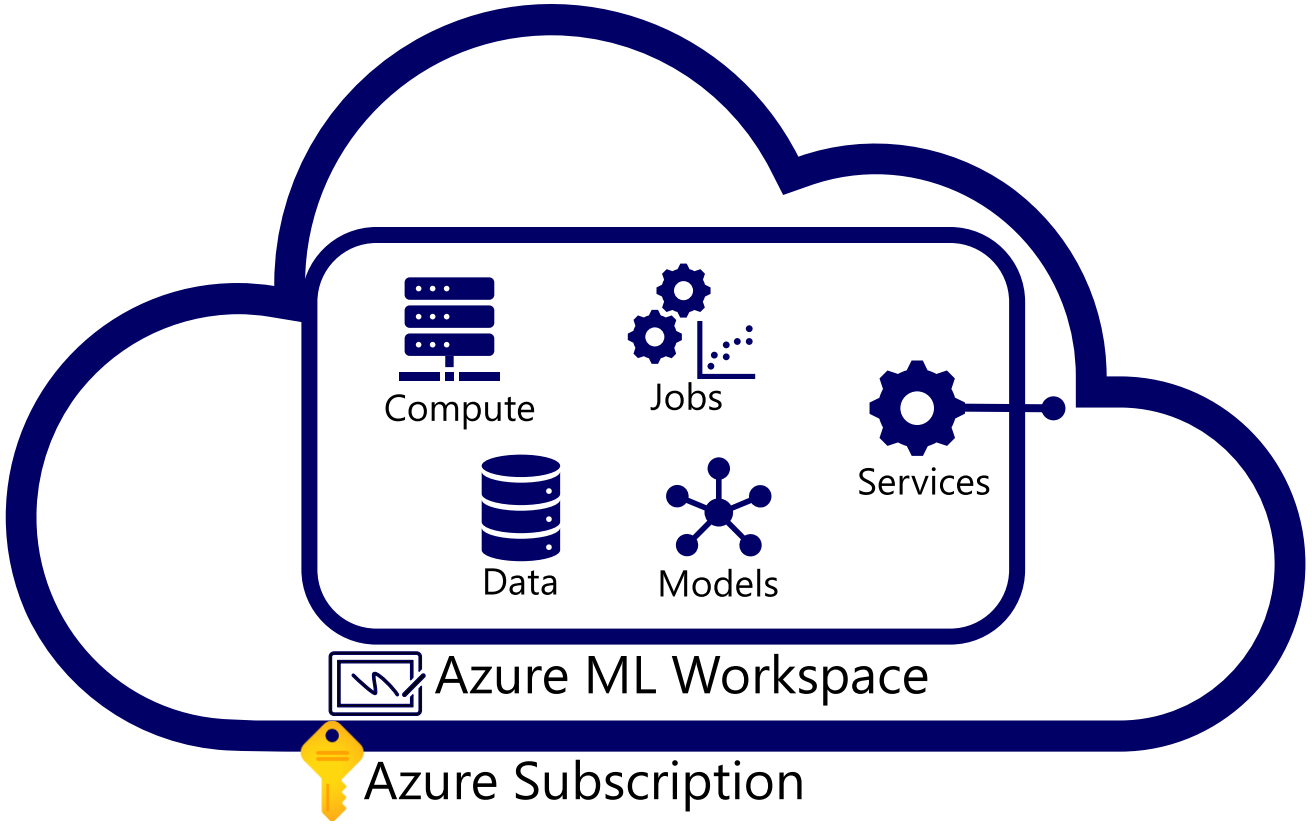


1. Split the data into a *training* set and a *validation* set
2. Apply an algorithm to *fit* the training data to a model
3. The trained model encapsulates the relationships in the data
4. Use the model to generate predictions from the validation data
5. Use evaluation metrics to compare predicted vs actual labels (supervised) or measure cluster separation (unsupervised)
6. Repeat…

Lesson 2: Azure Machine Learning

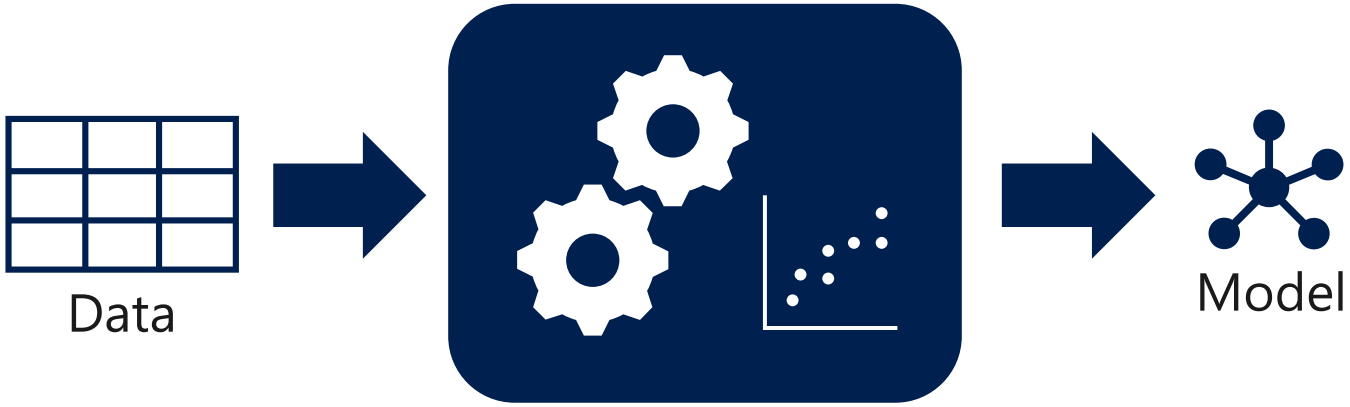
What is Azure Machine Learning?

A cloud-based platform for machine learning



Automated Machine Learning

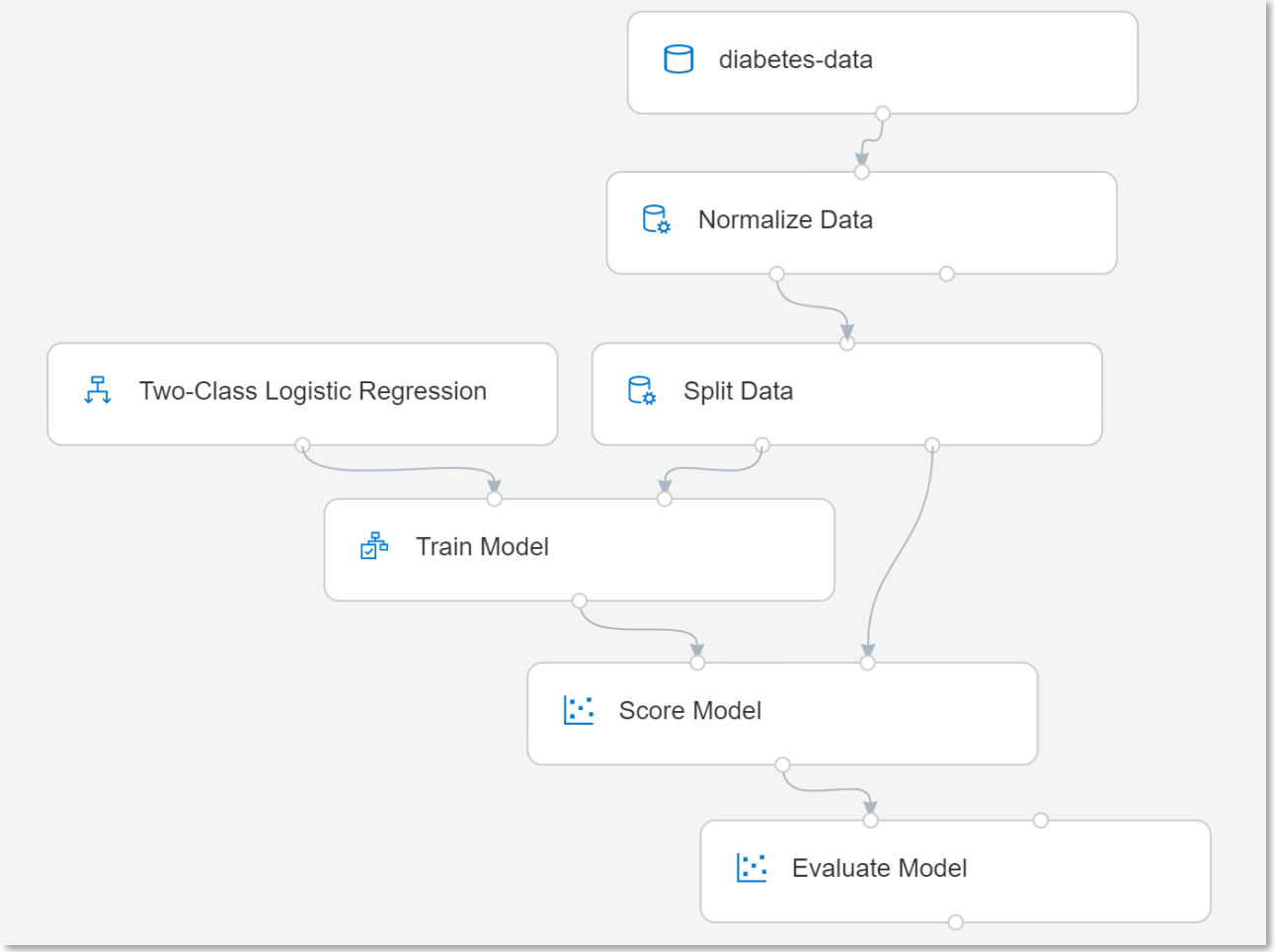
Supply the data and desired *supervised* model type, and let Azure Machine Learning find the best model



Azure Machine Learning Designer

Visual tool for creating a machine learning *pipeline*

1. Use a *training pipeline* to train and evaluate a model
2. Create an *inference pipeline* to predict labels from new data
3. Deploy the inference pipeline as a *service* for apps to use



Demo

Explore Automated Machine Learning

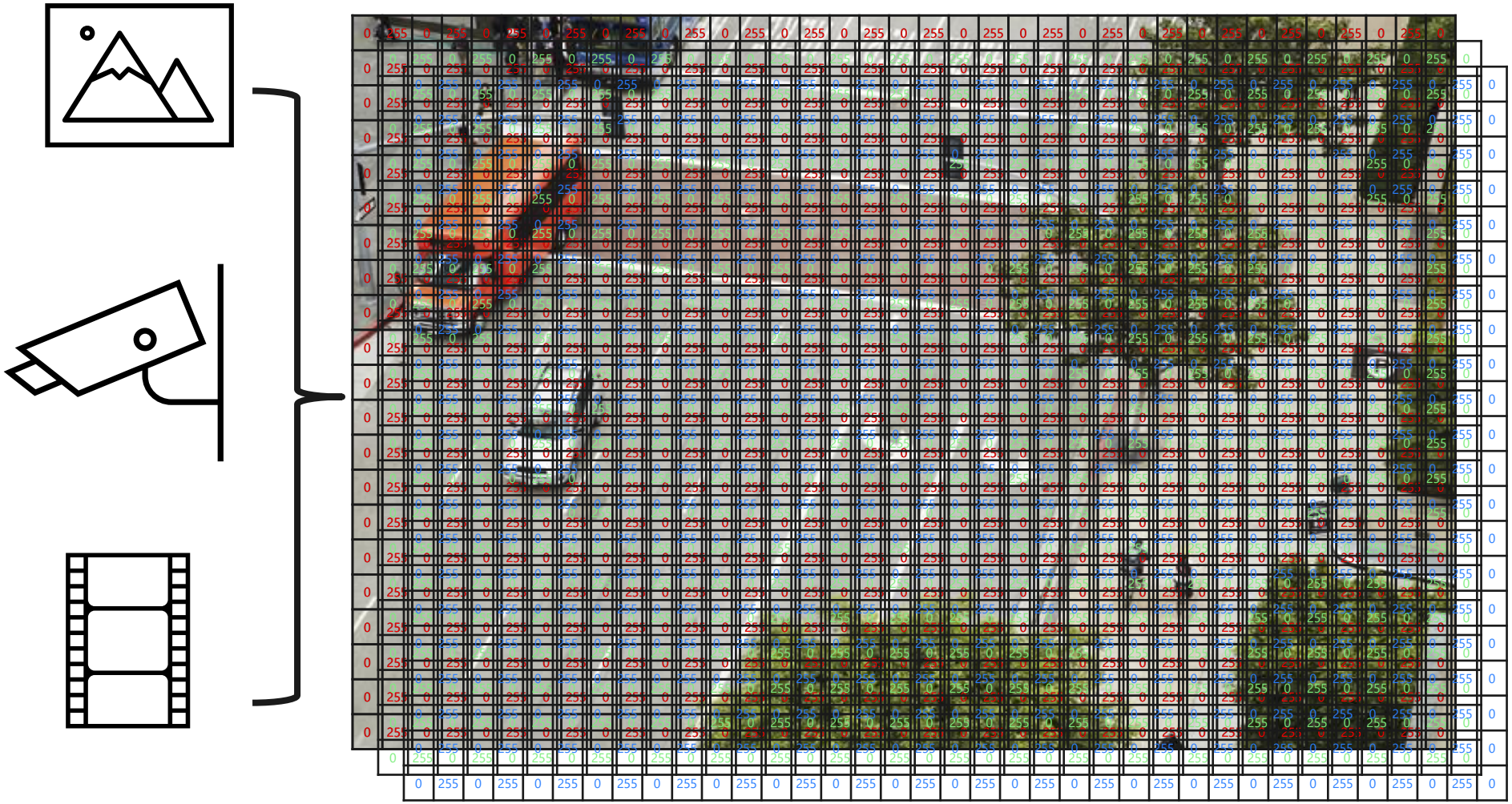
Module 3:

Explore Fundamentals of Computer Vision

- Lesson 1: Computer Vision concepts
- Lesson 2: Creating Computer Vision solutions in Azure

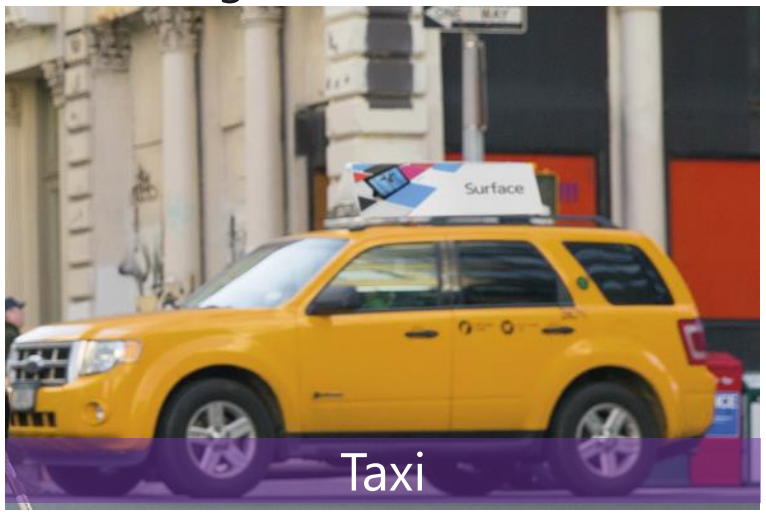
Lesson 1: Computer Vision concepts

What is Computer Vision?

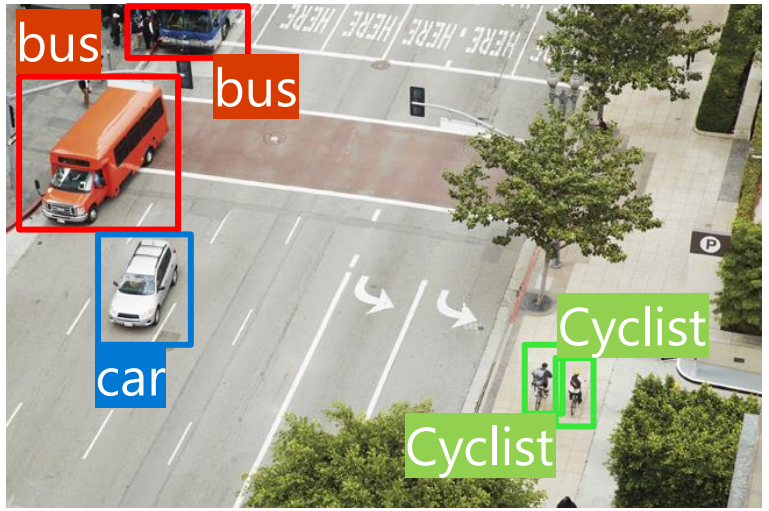


Applications of Computer Vision

Image Classification



Object Detection



Semantic Segmentation

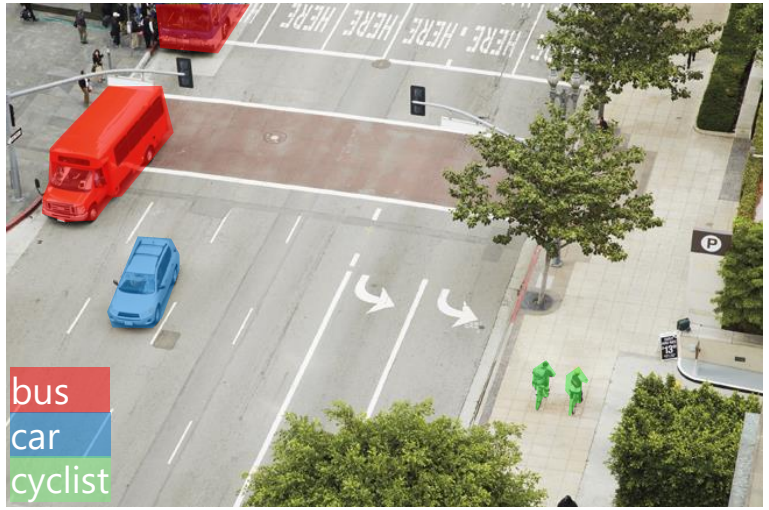
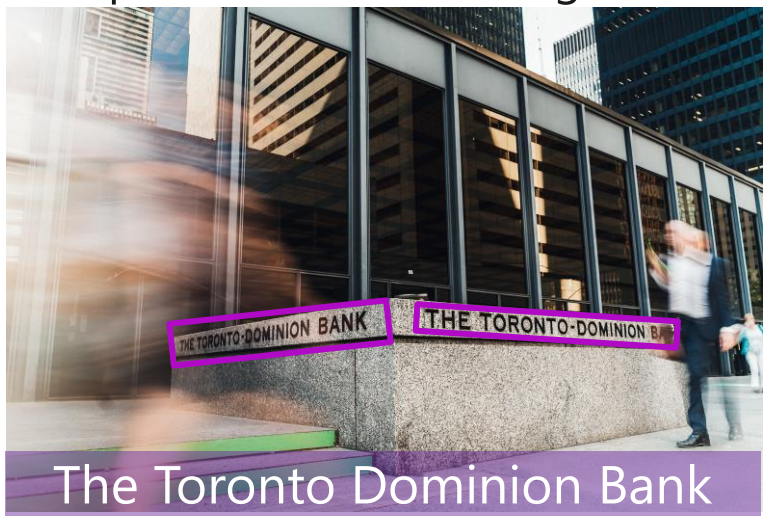


Image Analysis

Face Detection & Recognition

Optical Character Recognition



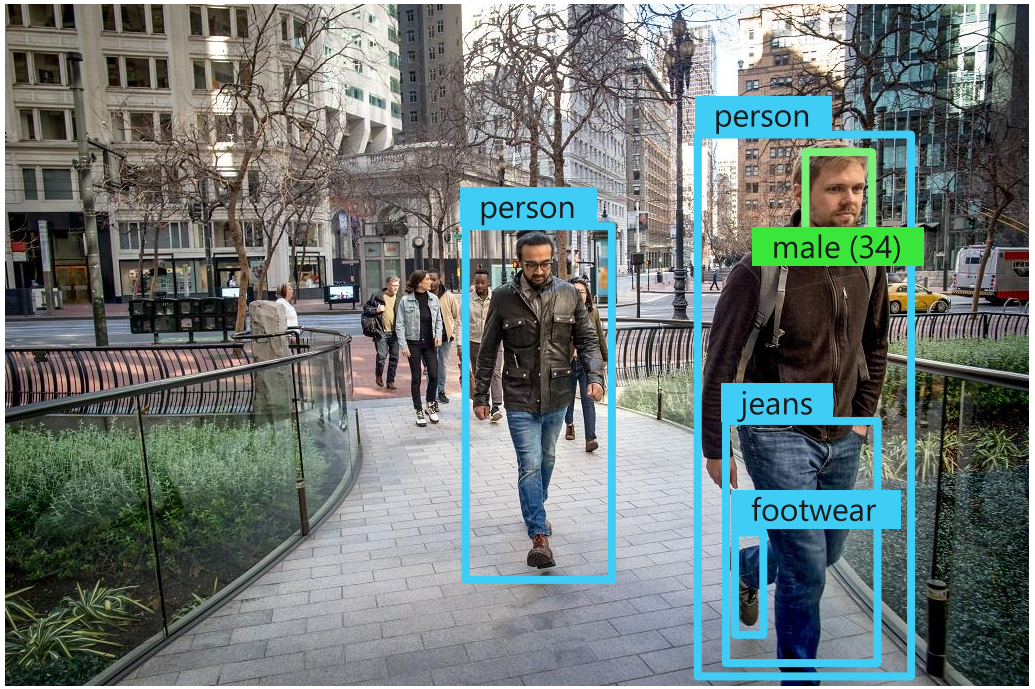
Computer Vision Services in Azure

Computer Vision	• Image analysis – automated captioning and tagging • Common object detection • Face detection • Smart cropping • Optical character recognition
Custom Vision	• Custom image classification • Custom object detection
Face	• Face detection and analysis
Form Recognizer	• Data extraction from forms, invoices, and other documents

Lesson 2: Creating Computer Vision solutions in Azure

Image Analysis with the *Computer Vision* Service

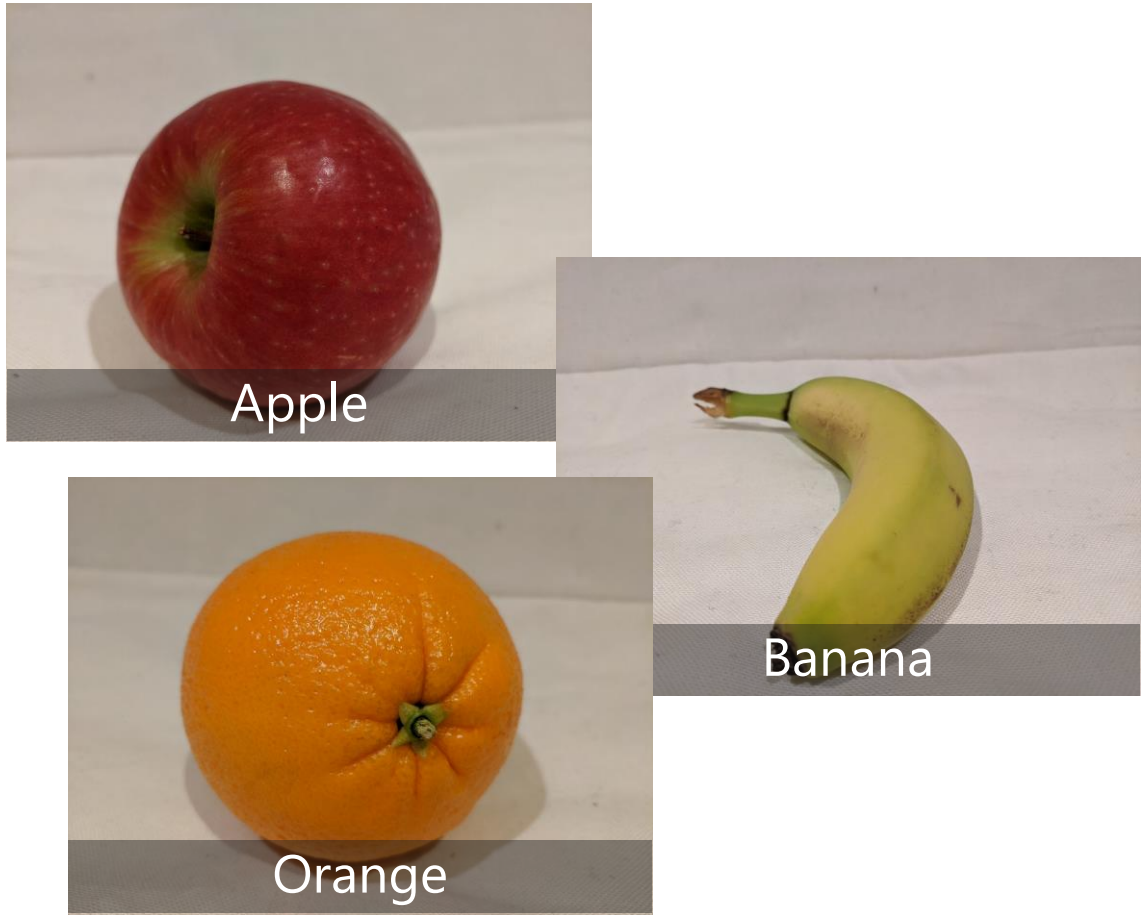
- Pre-trained computer vision model
- Object detection for over 10,000 predefined classes
- Image description and tag generation
- Face detection and analysis
- Content moderation
- Text detection and OCR



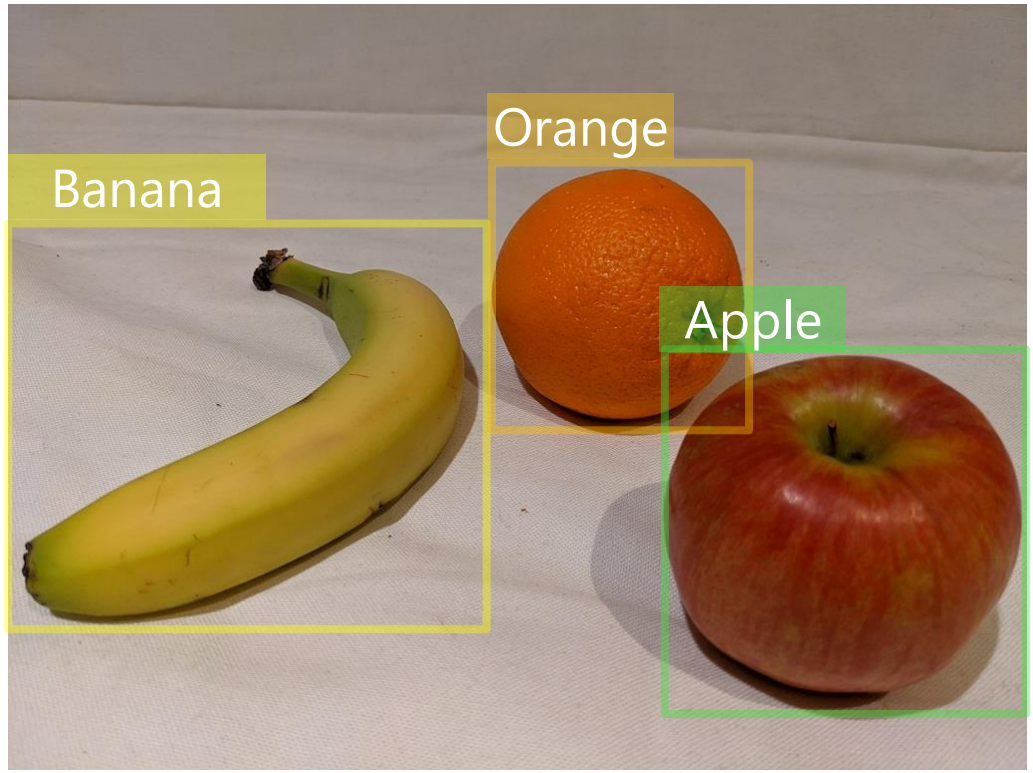
Caption: a group of people walking on a sidewalk
Tags: building, jeans, street, outdoor, jacket, city, person
Ratings: *Adult*: False, *Racy*: False, *Gore*: False

Training Models with the *Custom Vision* Service

Image Classification



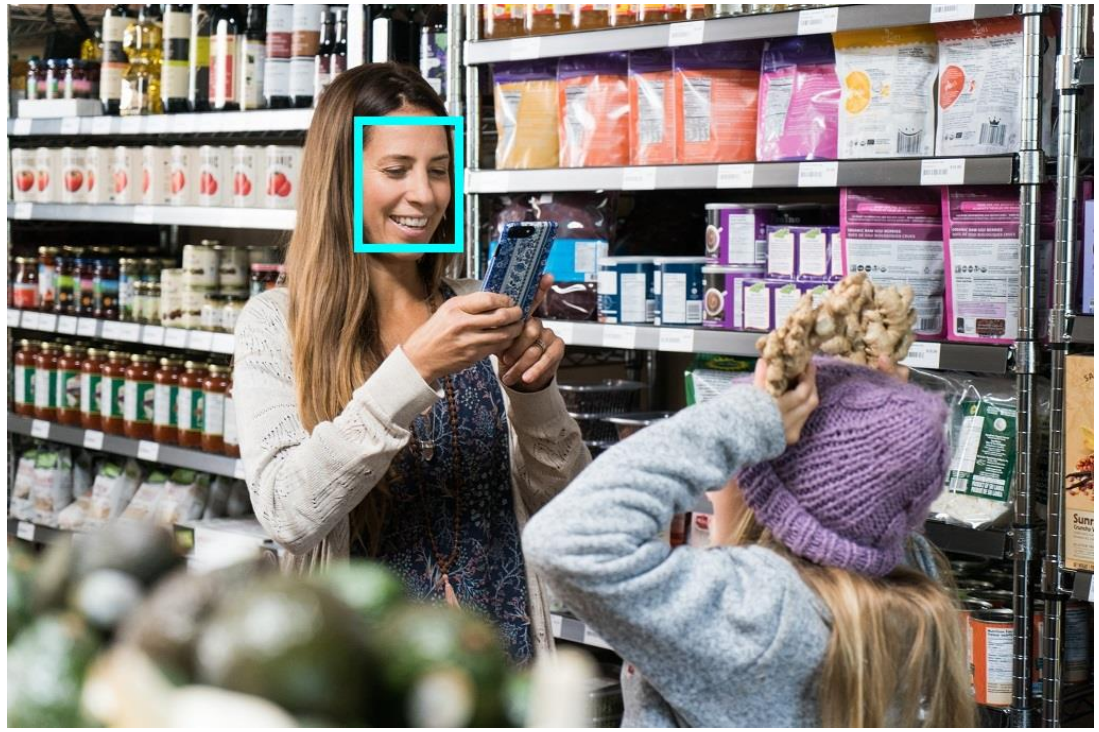
Object Detection



Detecting Faces with the *Face* Service

Anyone can use the Face service to detect:

- Blur
- Exposure
- Glasses
- Head pose
- Noise
- Occlusion



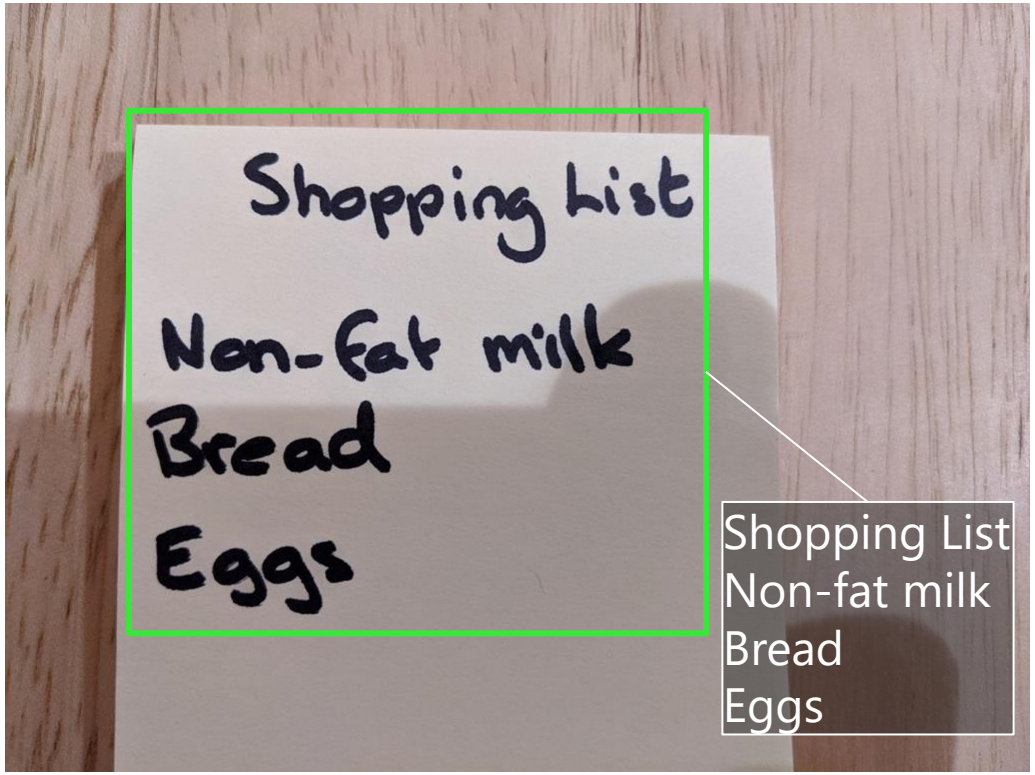
Only Managed Microsoft customers can access facial recognition capabilities:

- Similarity matching
- Identity verification

*To support Microsoft's Responsible AI Principles, Facial Recognition is under a Limited Access policy.

Reading Text with the *Computer Vision* Service

- Detect the location of text:
  - Printed
  - Handwritten
- Options for quick text extraction from images, or asynchronous analysis of larger scanned documents



Analyzing Forms with the *Form Recognizer* Service

- Extract information from scanned forms in image or PDF format
  - Use the pre-trained models for common document types
  - Train a custom model using your own forms
- Models perform *semantic recognition* of form fields – not just text extraction

Demo

Analyze images with Computer Vision

Module 4:

Explore Fundamentals of Natural Language Processing

- Lesson 1: Introduction to natural language processing
- Lesson 2: Building natural language solutions in Azure

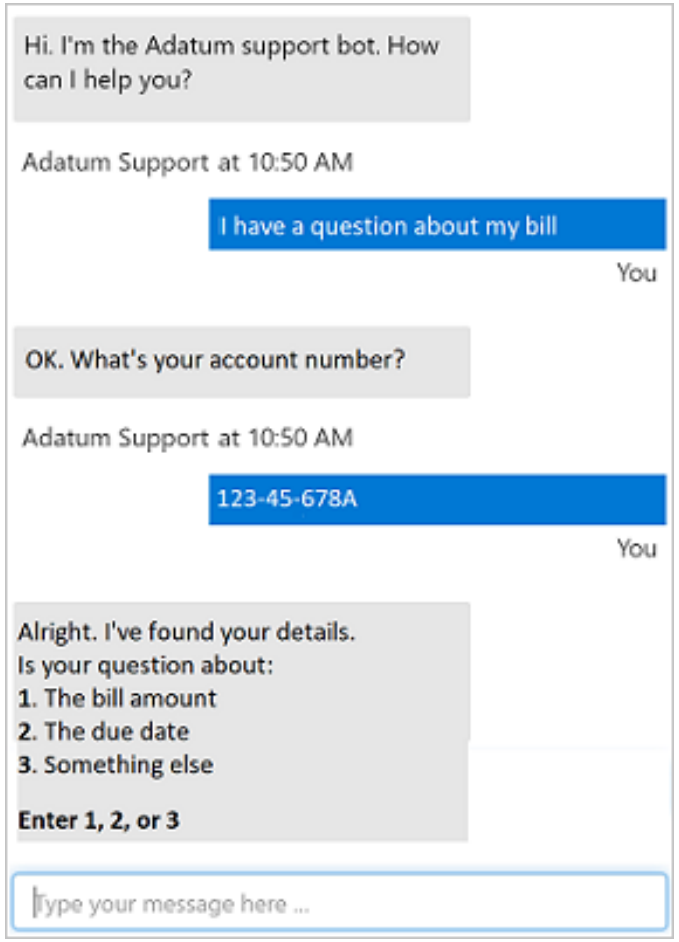
Lesson 1: Introduction to natural language processing

What is Natural Language Processing?

- Text analysis and entity recognition
- Sentiment analysis
- Speech recognition and synthesis
- Machine translation
- Semantic language modeling

What is Conversational AI?

- A solution that enables a dialog between an AI agent and a human
- Generically, conversational AI agents are known as *bots*
- Bots can engage over multiple *channels*:
  - Web chat interfaces
  - Email
  - Social media platforms
  - Voice



Natural Language Processing and Conversational AI in Azure

Language	• Language detection • Key phrase extraction • Entity detection • Sentiment analysis • Question answering • Conversational language understanding
Speech	• Text to speech • Speech to text • Speech translation
Translator	• Text translation
Azure Bot Service	• Platform for conversational AI

Lesson 2: Building natural language solutions in Azure

Analyzing Text

I had a wonderful vacation in France.

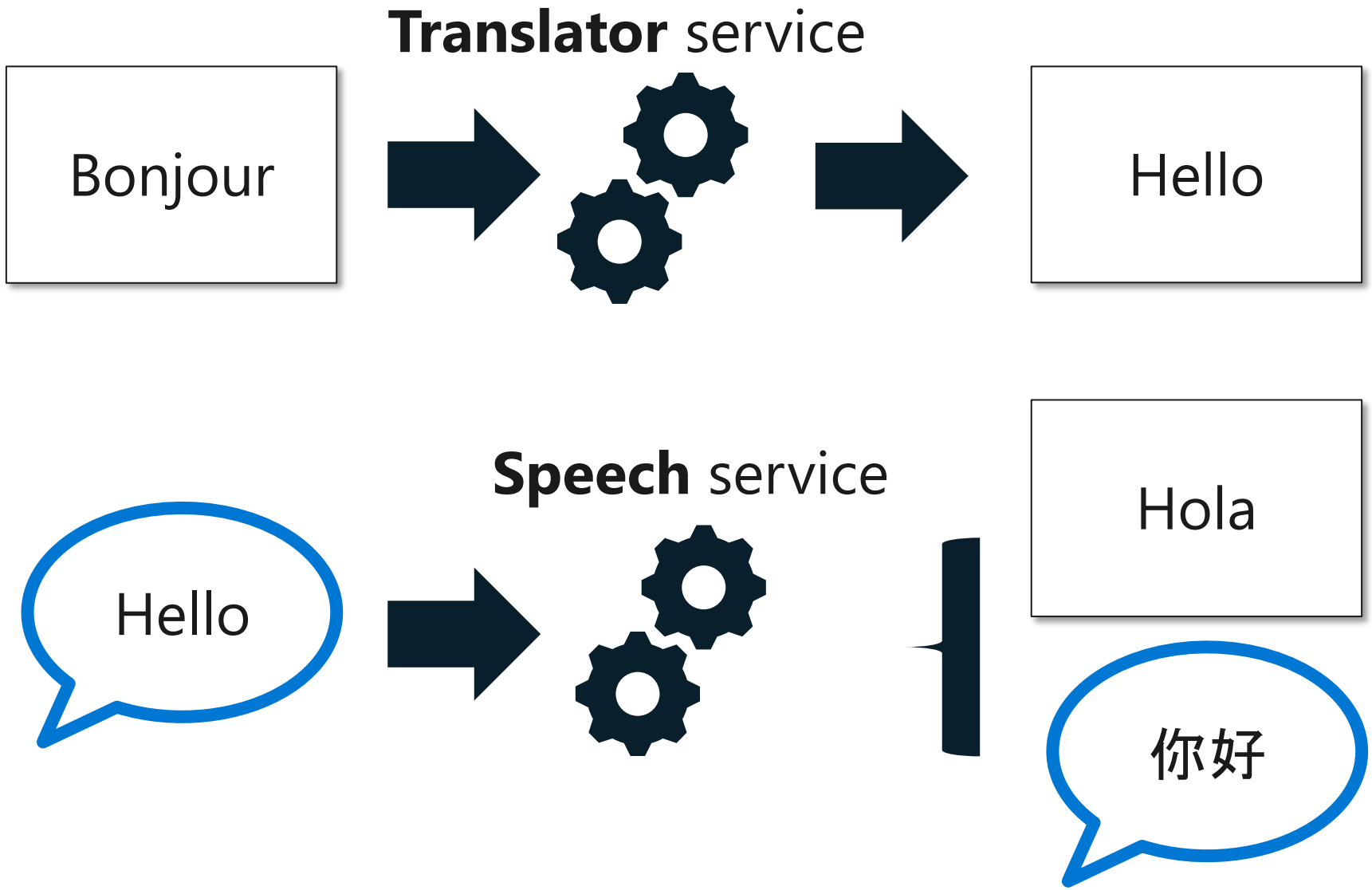
- Predominant Language: English
- Sentiment: 88% (positive)
- Key Phrases: "wonderful vacation"
- Entities: France

Speech Recognition and Synthesis

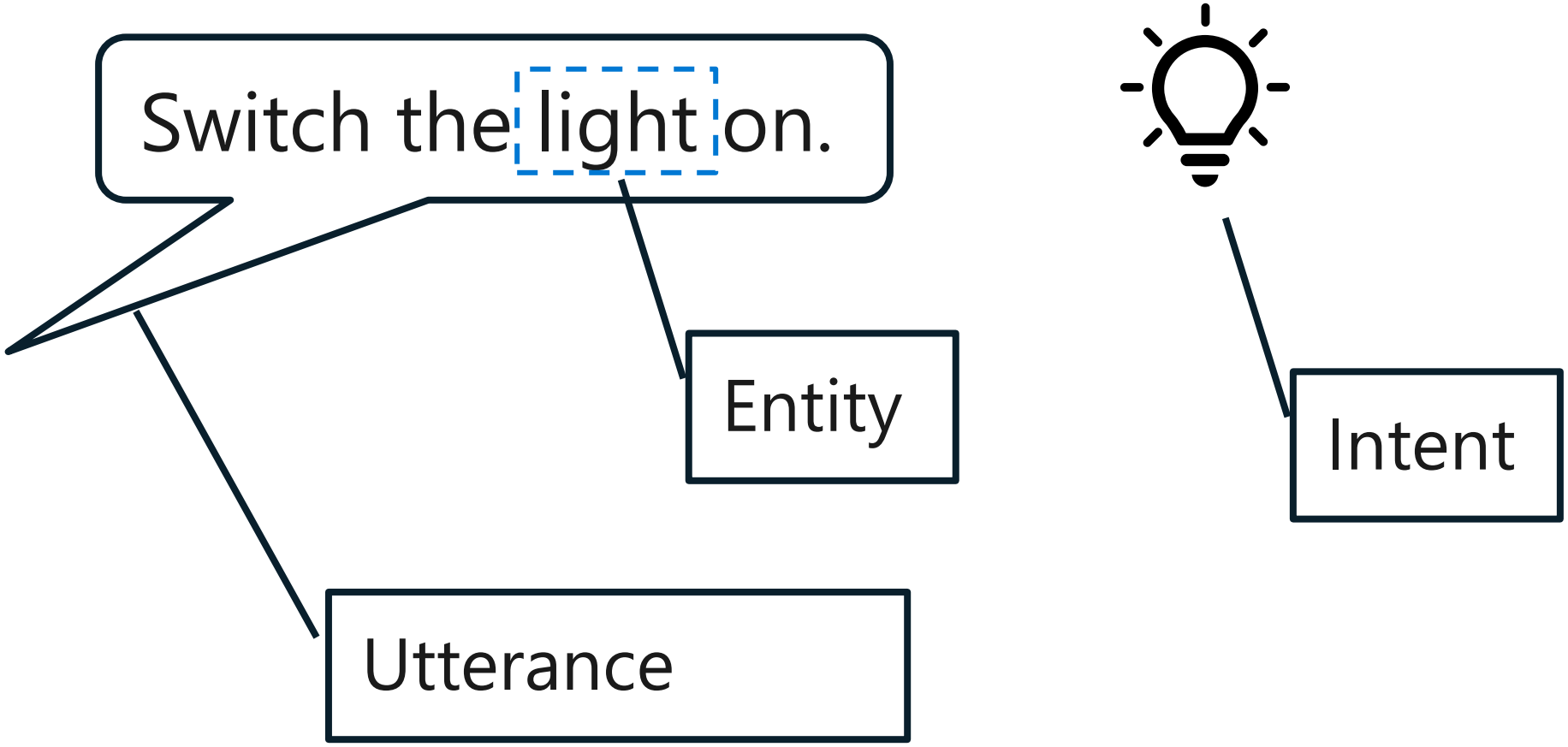
Use the *speech-to-text* capabilities of the **Speech** service to transcribe audible speech to text

Use the *text-to-speech* capabilities of the **Speech** service to generate audible speech from text

Translation

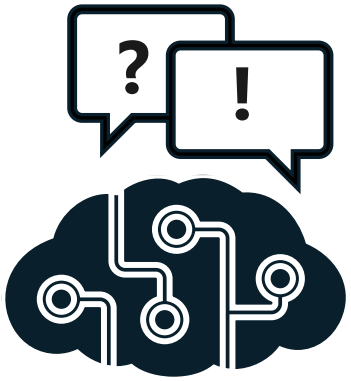


Conversational Language Understanding

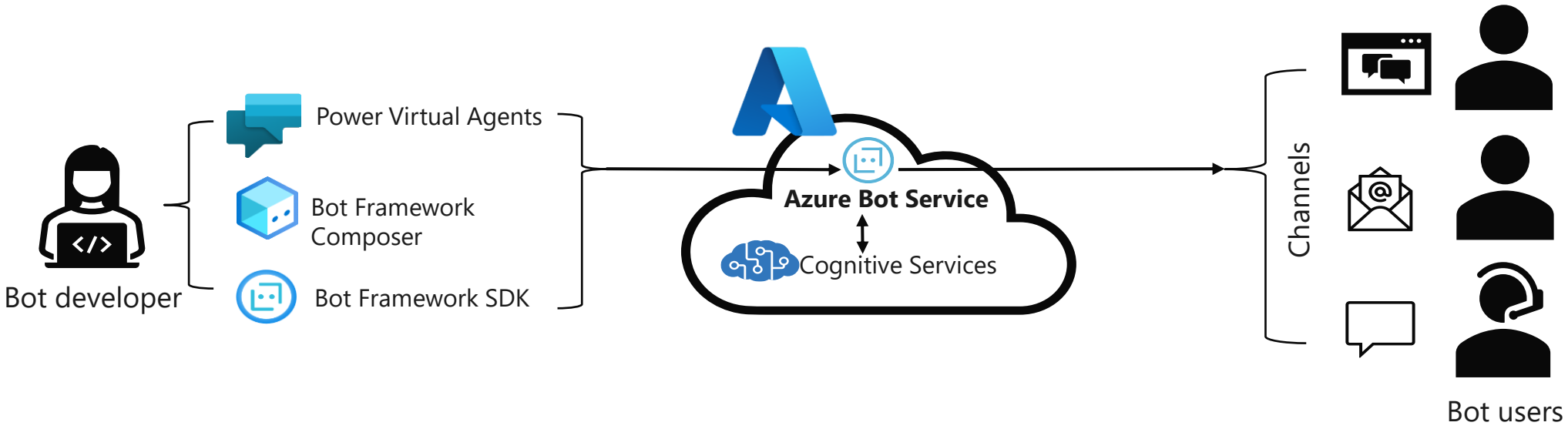


Question Answering

- Define a *knowledge base* of question and answer pairs:
  - By entering questions and answers
  - From an existing FAQ document
  - By using built-in *chit-chat*
- The knowledge base is consumed by client applications and bots



Azure Bot Service



Create bots using multiple tools:

- Power Virtual Agents
- Bot Framework Composer
- Bot Framework SDK

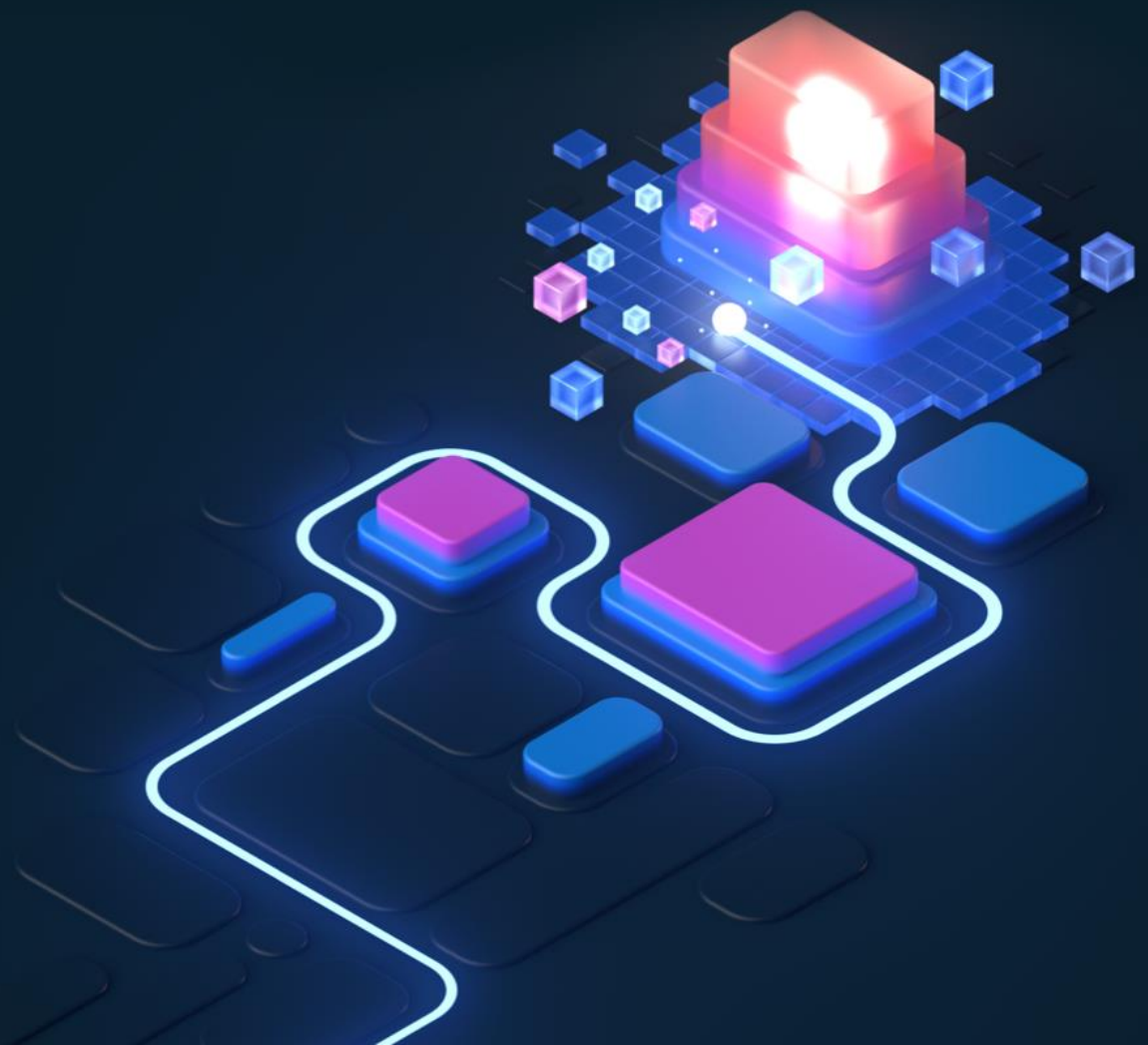
Use the Azure Bot Service to publish bots in the cloud and Integrate with cognitive services

Users interact with bots through multiple channels such as:

- Web chat
- Email
- Telephone

Demo

Analyze text with the Language Service

Microsoft

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